generally produces a more synchronized estrus than that obtained with a progestin-PMSG treatment, but subsequent fertility has been reported to be somewhat reduced. As stated previously for the ewe, many of these synchronization methods use products designed for cattle and other species; therefore, their use in the doe goat is considered an "off-label" use and should be used in consultation with a veterinarian or with knowledge of the appropriate regulatory guidelines.

18–2 Superovulation and Embryo Transfer

Artificial insemination makes it possible to utilize the sperm produced by the male reasonably efficiently. The potential ova contained in the ovary of the female have not been utilized effectively. Superovulation is a procedure in which the female is treated with hormones to cause her to produce several ova instead of the one that she normally produces at each estrus (Figure 18–3). In the ewe, doe, or cow, an average of 12 ovulations can be expected. In one study with 40 donor cows, an average of 7.9 embryos were collected nonsurgically. From 60% to 70% of superovulated ova form normal embryos. Reliable superovulation has not yet been achieved in the mare because ovulation occurs at one site on the ovary, the ovarian fossa.

Superovulation in calves has been accomplished and when made practical could shorten the generation interval by a year or more. Superior germ plasm could be identified earlier and utilized more efficiently. To date, the response has been variable and fertilization rates have been very low.

18–2.1 *Techniques of Superovulation*

The cow normally produces only 1 ovum per estrous period, the doe and ewe 1 to 3 ova (depending on breed), and the sow 12 to 20 ova per estrous period. The object of superovulation



Figure 18–3 Multiple ovulation sites can be seen on this superovulated ovary, which has been exposed by surgery. (Courtesy, of Codding Embryological Services, Foraker, Okla.)

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is to cause the female to produce a large number of ova that after fertilization and sufficient development can be transferred to other females.

Theoretically, superovulation can be accomplished at intervals equal to the length of the estrous cycle. With midluteal initiation of treatment, the interval may be shortened. Extreme variation between animals and diminished response with repeated treatments are some problems that still need to be solved. With current technology, donor cows are superovulated at 60-day intervals for an average of three ovulations. Some cows continue to respond for 10 treatments, but most become refractory much sooner. Allowing them to go through a pregnancy helps restore responsiveness. The reduced response may be due to an immune reaction in which antibodies are produced in response to the injections of the foreign proteins used on the donor to induce a superovulatory response. It may not be solved easily. Assuming that these problems can be overcome, up to 5 embryos from a cow might be obtained every 15 days. This would mean 120 or more embryos from one cow in a 12-month period.

18–2.1a Cows For a number of years, superovulation in cows was accomplished using FSHp, an extract obtained from the anterior pituitaries of pigs that had been recovered from slaughter plants. This hormone was injected twice daily for 4 to 5 consecutive days. A problem encountered when using FSHp was the variable response to a standard treatment. While some of this variability was due to responsiveness of the donor animal, a great deal of it was attributed to the purity of the FSHp. As an extract from the anterior pituitary, it contained varying amounts of LH and evidence was reported that ovulation rate decreased as the LH in the extract increased.

When this was confirmed, research was initiated to produce a FSH preparation with low LH activity by purifying the FSH extracted from the anterior pituitaries of pigs. This preparation of FSH with low LH:FSH activity has yielded a more consistent superovulation response than was previously possible with FSHp. A vial contains the FSH equivalent of 400 mg. Injections of FSH are started on day 8 to 10 following observed or induced estrus, and 50 mg are administered intramuscularly twice daily for 4 days (a total of eight injections of FSH).

Superovulation regimens can be started on any day of the estrous cycle between day 6 and day 15. However, one report indicated a poor response when started on day 11. For sake of superovulation response and ability to control the onset of estrus in the donor, most recommendations call for starting FSH on day 8, 9, or 10. If started on day 8, 9, or 10, the donor will have a corpus luteum to respond when $PGF_{2\alpha}$ is given at the time of the sixth FSH injection. With estrus in the donor controlled, recipients can be synchronized with the donors more easily. Progestin-based synchronization can be used in combination with a superovulation protocol, with injections of FSH traditionally started 5 or 6 days after progestin device insertion and $PGF_{2\alpha}$ given the day before or the day of progestin withdrawal. A summary of the steps used in superovulation of the donor using unpurified FSH extracts leading up to embryo transfer procedures appears in Table 18–4.

Some embryo transfer technicians prefer a regimen of descending doses of FSH during the 4-day injection period. For example, they might inject 60 mg twice on day 1, 50 mg per injection on day 2, 40 mg per injection on day 3, and 30 mg per injection on day 4. If a donor is not in estrus on day 5, 20-mg doses would be given. Three artificial inseminations have been recommended at 12-hour intervals starting at onset of estrus, when using the extracted FSH product containing LH. This was needed to provide adequate capacitated sperm at the site of fertilization to ensure maximum fertilization since sperm transport was impaired.

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Table 18–4 *Summary of events in superovulation and embryo transfer in the cow using unpurified FSH extracts*

Day of cycle	Action
9	Inject FSH at 7 A.M. and at 7 P.M.
10	Inject FSH at 7A.M. and at 7 P.M.
10	Inject recipients with 25 mg $PGF_{2\alpha}$ at 7 P.M.
11	Inject FSH at 7 A.M. and at 7 P.M.
11	Inject donor with 25 mg $PGF_{2\alpha}$ at 7 P.M.
12	Inject FSH at 7 A.M. and at 7 P.M.
13 (if not in estrus)	Inject FSH at 7 A.M. and at 7 P.M.
13–15	Estrus
Onset of Estrus	Inseminate
Onset of Estrus + 12 hours	Inseminate
Onset of Estrus + 24 hours	Inseminate
7 new cycle	Collect and evaluate embryos
7	Match embryos (stage of dev.) with recipients' day of cycle and transfer embryos into uterine horns of recipients
7	Inject $PGF_{2\alpha}$ into donor after the flush

The recent availability of recombinant FSH preparations for use in human-assisted reproduction has led to some investigations of such agents for use in cattle superovulation. However, studies suggest that heifers stimulated with recombinant human FSH, which possesses no LH bioactivity, in fact, require some LH activity in addition to FSH for normal preovulatory follicular development. Therefore, while low LH:FSH ratios are called for, exposure to both gonadotropins appears requisite.

Pregnant mare serum gonadotropin (PMSG) can also be used for superovulation in cattle and is given as a single dose of 2,000 IU to 2,500 IU. This is followed by a luteolytic dose of $PGF_{2\alpha}$ 2 to 3 days later. A second injection of $PGF_{2\alpha}$ is occasionally given 12 to 24 hours after the first and seems to improve embryo production in some cases. Experience has dictated that each batch (lot) of PMSG be tested for potency before it is used to superovulate a valuable donor cow. This may be done by injecting a grade cow with 2,000 IU as a starting point and adjusting the dose depending on the response. Use of FSH has surpassed that of PMSG as the primary method of superovulation in cattle. In most studies comparing the two procedures, the FSH treatment has resulted in slightly higher numbers of transferable embryos.

Dose level should be scaled to the size of the cow. If the dose level for a Holstein (700 kg) is 360 mg to 400 mg, the dose level for a Jersey (500 kg) should be only 300 mg to 340 mg. Rectal examination of ovaries during the injection period is recommended so that dose level can be reduced if overstimulation of ovaries is detected. Overstimulated ovaries may reach a diameter of 10 cm, while the desired size is 4 cm to 5 cm.

Donors selected for embryo transfer are cows that with special mating will produce offspring of high value. For good success, they should be less than 10 years old. Also, they should be cycling normally and be in good general and reproductive health. They should be in good body condition but not obese (Sections 23–3.1 and 23–3.2). With respect to other species of bovids, superovulation protocols for water buffalo are similar to the procedures described using FSH and PMSG, however some reports suggest that the

buffalo's ovarian response to FSH and PMSG are weak, compared with that domestic cows. Another major problem with superovulation in the water buffalo is the difficulty in detecting estrus for both donors and recipients (Section 7–2).

18–2.1b Sows and Gilts Because sows and cycling gilts normally ovulate 10 to 25 ova each cycle, superovulation in swine has not been essential for embryo production. However, evidence suggests that differences in ovulation rates of donors may affect pregnancy rates after transfer, with greater success from those with higher ovulation rates. In sows, 750 to 1,500 IU of PMSG (dose varies with size) should be injected subcutaneously on day 15 of the cycle. At the onset of estrus, 500 IU of hCG is injected subcutaneously. In cycling animals, PMSG can also be administered after synchronization of estrus with oral progestin (altrenogest) treatment. In a recent review of 12 studies using superovulation in gilts, some studies reported responses as high as 32 embryos collected; in some cases, no improvement in superovulatory response was observed, attesting to the highly variable superovulatory responses that may be seen even in animals of the same age, breed, and weight.

18–2.1c Ewes Although most ewes respond to superovulatory regimens, embryo recovery rates vary between approximately 30% to 60%. Six hundred to 1,000 IU of PMSG injected subcutaneously on day 12 or 13 of the cycle has been effective for increasing the number of ova ovulated in ewes. When used in combination with an estrous synchronization protocol, 1,000 IU of PMSG are injected 1 to 2 days prior to intravaginal progestin sponge removal. While the level of LH released from the ewe's pituitary gland may be adequate to cause ovulation of the induced follicles, 750 IU of hCG are usually given the day after progestin withdrawal to assure ovulation of all the follicles. In a recent study of St. Croix White ewes synchronized and superovulated with this protocol, control ewes that were only synchronized averaged 2.7 ovulations, while superovulated ewes averaged 6.7 ovulations. The administration of FSHp at the end of progestin treatment in decreasing doses has also yielded some success in the ewe, during both the breeding and nonbreeding (anestrus) seasons; however, further research is needed as to the appropriate protocols.

18–2.1d Does The superovulatory response in the doe is lower and more variable with PMSG than in a FSH-induced superovulation. Problems associated with PMSG-induced superovulation in the doe are an increased number of nonovulated follicles and short, irregular estrous cycles. PMSG is usually given as a single injection of 1,500 to 2,000 IU at the end of a progestin treatment. In contrast, FSH is administered in decreasing doses of 1 to 5 mg, injected at 12-hour intervals, over a 3- to 5-day period around the time of progestin withdrawal. Ovulation rates in does following FSH superovulation protocols can be as high as 10 to 25 ova, but the number of viable embryos may be significantly lower.

18–2.2 *Embryo Collection and Evaluation*

The history of embryo transfer began in 1890, when Walter Heape transferred embryos from an Angora rabbit to recipient Belgian rabbits. This experiment not only was the first successful embryo transfer but also demonstrated that the recipients' genetics would not un-

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duly influence the transferred embryos' genetic makeup or developmental processes. In 1930, the first embryo was collected from a cow, but it was not until 1951 that the first embryo transfer in the bovine was conducted using surgical means, and, in 1964, the first nonsurgical method of embryo collection was successful in cattle. Embryo transfers in sheep and goats were first reported in 1949; for the pig, in 1951. Offspring from frozen mouse embryos were produced in 1972, and the following year the production of offspring from frozen cow embryos were reported.

Early collection techniques involved either slaughtering the females and excising the oviducts or surgically removing the oviducts from the live females at 72 hours postovulation so that the embryos could be recovered by flushing. This defeated the primary purpose of superovulation, so other methods were developed. A surgical method was developed first and is accomplished by performing a *laparotomy* (flank or midline abdominal incision) to expose the reproductive tract (Figure 18–4). A clamp or the thumb and forefinger can be used to block the distal one-third of the uterine horn so that fluid injected into that segment can be forced through the oviduct with a gentle milking action and collected at the infundibulum. An alternate procedure is to occlude the uterine horn at the body of the uterus, with culture medium being introduced through a puncture at the uterotubal junction or through the oviduct until the uterus is turgid. The uterus is then punctured with a blunt needle attached to a flexible catheter. The pressure will cause the medium to gush through the catheter with enough turbulence to carry the embryos into a collection tube. These procedures allow for the recovery of a high percentage of embryos, but because of the surgical trauma and resulting adhesions it can be repeated only a few times. The adhesions make it difficult, if not impossible, to expose the reproductive tract repeatedly.

Nonsurgical techniques of recovery have been developed for the cow and mare that give results essentially equal to surgical methods. They involve the use of a size 18 to 24 French Foley catheter (two-way flow catheter), which allows flushing fluids to pass into the uterus and at the same time allows fluids to be returned from the uterus to a collecting



Figure 18–4 A surgical procedure is being used to flush embryos from the oviduct. (Courtesy of Codding Embryological Services, Foraker, Okla.)

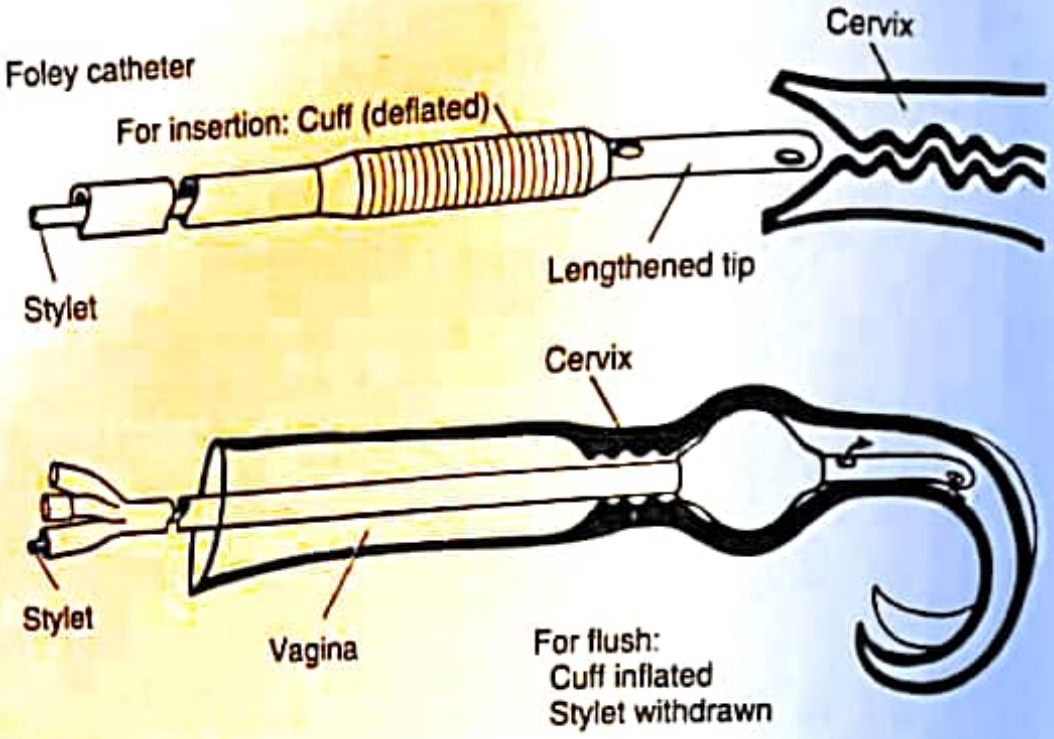


Figure 18–5 Foley catheter used to flush embryos from the uterus by nonsurgical means.

receptacle (Figure 18–5). A small balloon near the end of the catheter, which can be inflated just inside the uterine horn to prevent the flushing fluid from escaping through the cervix, is also a feature. The Foley catheter is larger in diameter than the usual insemination instrument. As embryo collections are conducted during early diestrus (days 6 to 8 after breeding), when the cervical lumen is constricted and lubricating mucus like that present during estrus is absent, the Foley catheter often cannot be passed through the cervix without the aid of a cervical dilator to physically open the cervical canal.

In sheep and goats, embryo collection and transfer has been hampered by the difficulty of introducing a catheter through the cervix and the inability to perform rectal manipulation of the reproductive tract. This has led to the use of predominately surgical means of embryo recovery in the past. Laparoscopic means, which results in fewer adhesions than traditional surgical techniques, have produced some success in the ewe and doe. One study investigating repeated embryo recoveries in subsequent estrous cycles in the ewe achieved embryo recovery rates ranging from 35% to 76% using laparoscopic approaches. With the development of improved catheters specifically for sheep and goats, some success has been achieved using nonsurgical, transcervical approaches. In does, studies have shown nonsurgical embryo recovery rates to be similar to those of surgical approaches, with one study demonstrating a 78% total embryo recovery rate and 62% transferable embryo recovery rate. These studies, and others, indicate that less invasive and nonsurgical means are feasible in small ruminants for embryo collection and transfer.

With nonsurgical collection methods, it is difficult to determine how many ovulation sites are present on the ovary, so it is not possible to determine when all of the embryos have been collected. In controlled experiments, about 50% of embryos resulting from superovulation were recovered by either surgical or nonsurgical procedures. Embryos are recovered from donor animals in the morula or early blastocyst stage. In the cow, ewe and doe, best re-

sults are obtained when embryos are recovered between days 6 and 8 (day 0 equals estrus). Conception rates of recipients following transfer of embryos collected on day 6 or 7 can produce conception rates of 70% to 80%, whereas embryos collected and transferred on day 4 or 5 have lower conception rates of 50% to 60%. Embryos can be recovered earlier when they are still in the oviducts of the donor at around the eight-cell stage. These embryos need to be transferred into the oviducts of synchronized recipients or are cultured *in vitro* to the blastocyst stage (Section 19–12) when they can be transferred into the recipient's uterus.

The fluids for flushing must be compatible with the embryos. A number of fluids have been successfully used, but the ideal medium is yet to be discovered. Autoserum (from the same species) alone or with an equal volume of 0.9% saline solution has been recommended for ewes. All sera used for flushing or culture of embryo of any species must be heated to 55°C for 30 minutes to deactivate factors detrimental to embryos. For most, if not all species, Modified Dulbecco's phosphate-buffered medium can be used (Table 18–5). Antibiotics are added to protect the embryos from bacterial contaminants acquired during flushing and to avoid transferring pathogens to the recipient. The donor cows are usually treated with large doses of antibiotics after flushing. Injection of $PGF_{2\alpha}$ is recommended to speed recovery of the ovaries and to prevent pregnancy, if viable embryos are not dislodged by the flush. In addition, an injection of $PGF_{2\alpha}$ is given to speed recovery of the ovaries and to prevent pregnancy in the donor in the event viable embryos were left behind.

The success rate for embryo transfer depends a great deal on an accurate evaluation of the embryos recovered. Experience has shown that certain deviations from normal development in 6- to 8-day embryos can be detected with the light microscope. Some characteristics which should be observed in an evaluation process are (1) compactness of the

Table 18–5 *Modified Dulbecco's phosphate-buffered medium*

Ingredients	Amount for 10 liters g
Part I	
NaCl	80.0
KCl	2.0
Na_2HPO_4	11.5
KH_2PO_4	2.0
Glucose	10.0
Streptomycin sulphate	.5
Na pyruvate	.36
Na penicillin G	1,000,000 units
Part II	
$CaCl_2\,2H_2O$	1.32
$MgSO_4\,7H_2O$	1.21

1. Dissolve part I in 8 liters of deionized, distilled water.
2. Dissolve part II in 2 liters of deionized, distilled water.
3. Add part II to part I slowly with constant stirring to prevent precipitation.
4. Add heat-treated autoserum immediately prior to use: 1% for flushing and 20% for culture.
5. Sterilize by passing through millipore filters with 0.22μ pores.

Adapted from Seidel et al. General Series 975. Colorado State Univ., Exp. Sta. and Animal Reprod. Lab. 1980.

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Figure 18–6 An unfertilized sheep ovum.

cells—a normal embryo is compact rather than a loose mass of cells; (2) regularity of shape—spherical is preferable to oval, etc.; (3) variation in cell size— blastomeres of similar size; (4) color and texture of cytoplasm—should be neither very light nor very dark; (5) presence of vesicles—cytoplasm should not be granular or unevenly distributed and should contain some moderate-sized vesicles; (6) presence of extruded cells—there should be no disassociated cells; (7) normal embryo size; (8) regularity of the zona pellucida—perivitelline space should be empty and of regular diameter, and the zona pellucida even (neither wrinkled nor collapsed); (9) presence of cellular debris—no cell fragments should be visible around the blastomeres. Using these subjective criteria, embryos are classified as Grade 1: excellent or good; Grade 2: fair; Grade 3: poor; or Grade 4: dead or degenerating. Embryos are also evaluated for their stage of development as follows: Stage 1: unfertilized (Figure 18–6); Stage 2: 2 to 12 cells; Stage 3: early morula; Stage 4: morula; Stage 5: early blastocyst; Stage 6: blastocyst; Stage 7: expanded blastocyst; Stage 8: hatched blastocyst; and Stage 9: expanded hatched blastocyst. Figure 9–1 shows embryos at different stages of development. In a recent study of beef and dairy cows composed of 15 different breeds, 58% of all embryos collected were considered transferable, 31% were unfertilized, and 11% were degenerated.

18–2.3 *Selection of Recipients and Transfer of Embryos*

Another important factor in successful embryo transfer is selection of recipients. Recipients should be young, reproductively sound, and cycling normally. If heifers are used, they should meet size standards for breeding (Section 20–2.1) because small heifers will increase incidence of dystocia and neonatal loss (Section 10–5). Good body condition and good general health are important. Their cycles should be synchronized with that of the donor and they should have a well-formed corpus luteum at the time of transfer. Failure to provide recipients that meet these criteria is a frequent shortcoming in countries that do not have an established embryo transfer program. Some report that *Bos indicus* (Zebu) cattle have lower embryo survival after transfer than *Bos taurus* or *Bos indicus/Bos taurus* crosses.

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Embryos must be transferred to recipients that have their estrous cycles synchronized with that of the donor. Recipients that are 2 days ahead or 2 days behind the day of the cycle of the donor when embryos are collected and transferred can reduce pregnancy rates by 20% to 50% in sheep and cattle. Techniques for estrous synchronization were discussed in an earlier section of this chapter (Section 18–1.1). When recipients are synchronized with PGF_{2a}, they should be injected three days before the anticipated estrus of the donor or 12 hours before injecting the donor with PGF_{2a}. This optimizes the probability that the recipient will be at the same stage of the estrous cycle as the donor when the transfer is performed. If using a progestin implant, removal should be 2 days before the donor's estrus is expected. Both surgical and nonsurgical techniques have been used for transferring embryos. The surgical procedure involves a laparotomy to expose the reproductive tract. The normal embryo, depending on stage of development, is placed in either the oviduct or the uterus of the recipient. A small syringe fitted with a 21-gauge needle is used to make the transfer. When the embryo is placed in the uterus, the needle is carefully inserted through the wall of the uterine horn adjacent to the ovary which contains the corpus luteum. When the embryo is placed in the oviduct, the needle is carefully inserted through the infundibulum into the ampulla where the embryo is deposited.

Two nonsurgical procedures for transferring embryos to recipients have been described and used in the cow and mare. In one procedure, a long hypodermic needle is inserted into the vagina and used to bypass the cervix and puncture the uterine horn so that the embryo can be deposited into the lumen of the uterus. Currently, the most common procedure is to place the embryo in a 0.25-ml plastic straw which has been described for packaging semen. This straw is then placed in the stainless steel embryo transfer gun, which is passed through the cervix for the deposit of the embryo into the uterus. The straw gun is carefully passed into the uterine horn adjacent to the ovary with the functional CL. Extreme caution should be taken to prevent injury to the endometrium. Nonsurgical transfers have a success rate somewhat lower than surgical methods; however, most commercial embryo transfers are made by nonsurgical means. A summary of the steps used in superovulation and embryo transfer in the cow using unpurified FSH extract appeared in Table 18–4.

18–2.4 *Storage, Freezing and Thawing Embryos*

Embryos can be maintained at near body temperature in the media used for flushing during the period between recovery and transfer to the recipients. If embryos are to be held longer than 2 hours before transfer, media containing 20% heat-treated serum should be used. The embryos should be transferred to fresh, sterile medium every 2 hours to help control bacterial contaminants. If embryos are cooled to 5°C (refrigerator temperature), they can be maintained for 2 to 4 days. Holding temperatures should be maintained and not allowed to fluctuate.

Embryos may be held in 20% heat-treated serum up to 10 hours without a reduction in survival. Lower temperature must be used if embryos are to be retained for longer periods of time. Sheep embryos held at 10°C have retained viability up to 72 hours. Embryos from cows and ewes may be frozen and stored in liquid nitrogen, but a 15 to 20 percentage units reduction in pregnancy rate should be expected.

Technology for freezing and thawing of bovine embryos has improved greatly in recent years. As with semen, cryoprotectants are needed for acceptable survival. Until recently, glycerol was the cryoprotectant used most frequently for freezing embryos.

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Equilibration of the embryo in a holding medium containing glycerol permitted glycerol to replace water within the embryo. As ice crystals started forming, the salts in the liquid holding medium became more concentrated, with the resulting change in osmotic pressure drawing more water out of the embryo as glycerol moved in. The problem came after thawing. The embryo requires rehydration before being placed into a recipient. Since water diffuses through cell membranes more quickly than glycerol can diffuse out, rehydration had to be done in a stepwise manner. Otherwise, excessive swelling frequently damaged cell membranes, destroying the embryo. Rehydration was accomplished by transferring the embryo through several holding media, each with a lower concentration of glycerol, until sufficient rehydration had occurred. Adding sucrose to the media as an osmotic buffer permitted more direct rehydration but was not accepted well by the industry.

More recently, ethylene glycol has been used as a cryoprotectant when freezing embryos. Since ethylene glycol diffuses rapidly through cell membranes, as does water, direct rehydration of embryos within straws is possible. To accomplish this, the embryo is placed in 1.5 M ethylene glycol in the center column of a 0.25-ml straw with a column of phosphobuffered saline on either end of the straw. The columns are separated by an air bubble. The recommended ratio of ethylene glycol to phosphobuffered saline in the straw is 1:3. Freezing follows a stepwise procedure. The prepared straw is placed directly into an alcohol bath freezer (−7°C). The straw is then seeded by touching a metal rod, cooled by immersion in liquid nitrogen, to the straw at a location away from the embryo. After a 5-minute hold, the prepared straw is cooled at 0.5°C per minute to −35°C, held for 15 minutes, and then plunged into liquid nitrogen. Straws are thawed in a 30°C water bath and then transferred directly to recipients. Rehydration occurs during the transfer. Pregnancy rates of about 50% are equivalent to that achieved by other accepted methods. Only top-quality embryos freeze well. Therefore, lower-quality but acceptable embryos should be transferred as fresh embryos only.

The advances in this technology have made international shipment of embryos more practical and competitive with shipment of live animals. Advantages of shipping frozen embryos over shipping live animals include availability of better genetics, elimination of disease transmission with proper processing of embryos, and opportunity for savings in costs, especially when the comparative genetics are considered. A concern in a number of countries is the availability of suitable recipients (Section 18–2.3).

18–2.5 *Promoting Twinning*

Transfer of two embryos to each of a group of recipients will increase the twinning rates in that group of females and has become the method of choice to accomplish that goal. Modifications of the superovulation treatment discussed in Section 18–2.1 have been employed to induce twinning in cattle and in sheep. While the number of offspring per pregnancy can be increased, it is not possible to limit the number of ovulations to only two. When the treatment produces several ova, space limitations in the uterus usually result in embryonic mortality to reduce the number of fetuses to one, two, or three. Unfortunately, it does not always occur this way; instead, too many implantations frequently occur and later result in abortions.

While twinning is desirable in sheep, the disadvantages outweigh the advantages for twinning in cattle. Some of these are (1) short gestations accompanied by small, weak

calves; (2) high incidence of retained placentae; (3) freemartins result from heterosexual twins; (4) more stress on the cows—dairy cows produce 10% less milk in the lactation following the delivery of twins; (5) increased reproductive problems—one study reported that 50% of the cows producing twins failed to calve the next year; (6) increased incidence of cystic ovaries—anytime exogenous FSH, particularly in the form of PMSG, is used, one must expect an increase in the incidence of cystic ovaries; and (7) most beef cows do not produce enough milk for two calves. Based on work with sheep, a considerably higher level of nutrition is needed during late gestation to support fetal growth.

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